

```
1
2
3 Imports System
4 Imports System.IO.Ports
5
6 Public Class IssueBook
7
8     Dim receivedData As String = ""
9
10    Function ReceiveSerialData() As String
11        receivedData = ""
12        Dim Reader As String
13        Try
14
15            Reader = SerialPort2.ReadExisting()
16            If Reader Is Nothing Then
17                Return "Nothing" & vbCrLf
18            Else
19                Return Reader
20            End If
21        Catch ex As TimeoutException
22            Return "Error"
23        End Try
24
25
26    End Function
27
28    '////////////////////////////////////
29
30    Function ArduinoConnect() As String
31        Timer2.Enabled = False
32
33        For Each sp As String In My.Computer.Ports.SerialPortNames
34            SerialPort2.Close()
35            SerialPort2.PortName = sp
36            SerialPort2.BaudRate = 9600
37            SerialPort2.DataBits = 8
38            SerialPort2.Parity = Parity.None
39            SerialPort2.StopBits = StopBits.One
40            SerialPort2.Handshake = Handshake.None
41            SerialPort2.Encoding = System.Text.Encoding.Default 'very ?
42                important!
43            SerialPort2.ReadTimeout = 10000
44            SerialPort2.Open()
45            Timer2.Enabled = True
46        Next
47    End Function
48
49    Private Sub IssueBook_Load(ByVal sender As System.Object, ByVal e As ?
50        System.EventArgs) Handles MyBase.Load
51        MetroTextBox8.Select()
52        MetroDateTime1.MinDate = DateTime.Today
53        MetroDateTime1.Enabled = False
54        MetroDateTime2.MinDate = DateTime.Today
55        Call readData()
56        MetroDateTime2.Value = MetroDateTime2.Value.AddDays(3)
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55     ArduinoConnect()
56 End Sub
57
58
59
60 Private Sub Timer2_Tick(sender As Object, e As EventArgs) Handles Timer2.Tick ↗
61     receivedData = ReceiveSerialData()
62     MetroTextBox9.Text &= receivedData
63 End Sub
64
65
66
67 Private Sub metrobutton4_Click(ByVal sender As System.Object, ByVal e As System.EventArgs) ↗
68     Me.Close()
69     LibraryMainForm.Show()
70 End Sub
71
72 Private Sub MetroTextBox8_KeyPress(sender As Object, e As KeyPressEventArgs) Handles MetroTextBox8.KeyPress ↗
73     'condition for entering number only
74     If Asc(e.KeyChar) <> 13 AndAlso Asc(e.KeyChar) <> 8 AndAlso Not ↗
75         IsNumeric(e.KeyChar) Then
76         MessageBox.Show("Please enter numbers only")
77         e.Handled = True
78     End If
79 End Sub
80
81
82
83 Sub readData()
84     MetroListView1.Clear()
85     MetroListView1.Columns.Add("Book Barcode", 160, ↗
86         HorizontalAlignment.Center)
87     MetroListView1.Columns.Add("Group ID", 60, HorizontalAlignment.Center) ↗
88     MetroListView1.Columns.Add("Book Name", 310, ↗
89         HorizontalAlignment.Center)
90     MetroListView1.Columns.Add("Publisher", 90, ↗
91         HorizontalAlignment.Center)
92     MetroListView1.Columns.Add("Author", 90, HorizontalAlignment.Center)
93     MetroListView1.Columns.Add("Publishing Year", 190, ↗
94         HorizontalAlignment.Center)
95     MetroListView1.Columns.Add("Edition", 90, HorizontalAlignment.Center)
96     MetroListView1.Columns.Add("Quantity", 90, HorizontalAlignment.Center)
97     MetroListView1.View = View.Details
98
99 Try
100
101     If (objcon.State = ConnectionState.Closed) Then objcon.Open()
102     com = New OleDb.OleDbCommand("SELECT * FROM Books", objcon)
103     dr = com.ExecuteReader
104     While dr.Read()
105         Call adddatatolistview(MetroListView1, dr(0), dr(1), dr(2), dr ↗
106             (3), dr(4), dr(5), dr(6), dr(7))

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102         End While
103         dr.Close()
104         objcon.Close()
105     Catch
106         'MsgBox("Please Refresh", MsgBoxStyle.Information, "")
107     End Try
108 End Sub
109 Public Sub adddatatolistview(ByVal lvw As ListView, ByVal BookID As String,
110                             ByVal GroupID As String, ByVal BookName As String,
111                             ByVal Publisher As String, ByVal Author As String,
112                             ByVal PubYear As String,
113                             ByVal edi As String, ByVal qty As String)
114     Dim lv As New ListViewItem
115     lvw.Items.Add(lv)
116     lv.Text = BookID
117     lv.SubItems.Add(GroupID)
118     lv.SubItems.Add(BookName)
119     lv.SubItems.Add(Publisher)
120     lv.SubItems.Add(Author)
121     lv.SubItems.Add(PubYear)
122     lv.SubItems.Add(edi)
123
124     lv.SubItems.Add(qty)
125 End Sub
126
127 Private Sub MetroTextBox8_SelectedIndexChanged(ByVal sender As
128     System.Object, ByVal e As System.EventArgs)
129     Dim i As Integer
130     MetroListView1.SelectedItems.Clear()
131
132     Try
133         If Me.MetroTextBox8.Text = "" Then
134             MetroTextBox6.Text = ""
135         Else
136             For i = 0 To MetroListView1.Items.Count - 1
137                 If MetroTextBox8.Text = MetroListView1.Items(i).SubItems
138                     (0).Text Then
139                     MetroTextBox6.Text = MetroListView1.Items(i).SubItems
140                     (1).Text
141                     MetroTextBox2.Text = MetroListView1.Items(i).SubItems
142                     (2).Text
143                     MetroTextBox3.Text = MetroListView1.Items(i).SubItems
144                     (3).Text
145                     MetroTextBox4.Text = MetroListView1.Items(i).SubItems
146                     (4).Text
147                     MetroTextBox7.Text = MetroListView1.Items(i).SubItems
148                     (5).Text
149                     MetroTextBox5.Text = MetroListView1.Items(i).SubItems
150                     (6).Text
151                     MetroTextBox12.Text = MetroListView1.Items(i).SubItems
152                     (7).Text
153                     MetroListView1.Items(i).Selected = True
154                     Exit For
155                 End If
156             Next
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146         End If
147     Catch
148
149     End Try
150 End Sub
151
152 Private Sub metrobutton3_Click(ByVal sender As System.Object, ByVal e As 
    System.EventArgs)
153     Try
154         If MetroTextBox8.Text = "" Then
155             MsgBox("Please mention the BookID", 0, "")
156         Else
157             If objcon.State = ConnectionState.Closed Then
158                 com = New OleDb.OleDbCommand("delete from Issue where 
159                 BookID='" & MetroTextBox8.Text & "'", objcon)
160                 If MsgBox("Do you really want to delete?", ,
161                 MsgBoxStyle.YesNo, "Are you sure?") = 
162                     Windows.Forms.DialogResult.Yes Then
163                         com.ExecuteNonQuery()
164                         Call readData()
165                     End If
166                 objcon.Close()
167             End If
168         End If
169     Catch ex As Exception
170
171     End Try
172 End Sub
173
174 Private Sub MetroListView1_SelectedIndexChanged(ByVal sender As 
    System.Object, ByVal e As System.EventArgs) Handles 
    MetroListView1.SelectedIndexChanged
175     Dim i As Integer
176     For i = 0 To MetroListView1.Items.Count - 1
177         If MetroListView1.Items(i).Selected = True Then
178             MetroTextBox8.Text = MetroListView1.Items(i).SubItems(0).Text
179             Exit For
180         End If
181     Next
182     MetroListView1.Focus()
183     MetroListView1.FullRowSelect = True
184 End Sub
185
186 Private Sub MetroTextBox8_TextChanged(sender As Object, e As EventArgs) 
    Handles MetroTextBox8.TextChanged
187     Dim i As Integer
188     MetroListView1.SelectedItems.Clear()
189     MetroTextBox8.Focus()
190     Try
191         If Me.MetroTextBox8.Text = "" Then
192             MetroTextBox6.Text = ""
193             MetroTextBox6.Text = ""
194             MetroTextBox2.Text = ""
195             MetroTextBox3.Text = ""

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195         MetroTextBox4.Text = ""
196         MetroTextBox7.Text = ""
197         MetroTextBox5.Text = ""
198
199         MetroTextBox12.Text = ""
200     Else
201         For i = 0 To MetroListView1.Items.Count - 1
202             If MetroTextBox8.Text = MetroListView1.Items(i).SubItems
203                 (0).Text Then
204                 MetroTextBox6.Text = MetroListView1.Items(i).SubItems
205                 (1).Text
206                 MetroTextBox2.Text = MetroListView1.Items(i).SubItems
207                 (2).Text
208                 MetroTextBox3.Text = MetroListView1.Items(i).SubItems
209                 (3).Text
210                 MetroTextBox4.Text = MetroListView1.Items(i).SubItems
211                 (4).Text
212                 MetroTextBox7.Text = MetroListView1.Items(i).SubItems
213                 (5).Text
214                 MetroTextBox5.Text = MetroListView1.Items(i).SubItems
215                 (6).Text
216                 MetroTextBox12.Text = MetroListView1.Items(i).SubItems
217                 (7).Text
218                 MetroListView1.Items(i).Selected = True
219                 Exit For
220             End If
221         Next
222     End If
223 Catch
224 End Try
225 End Sub
226
227
228
229 Private Sub MetroTextBox9_KeyPress(sender As Object, e As
230     KeyPressEventArgs) Handles MetroTextBox9.KeyPress
231     If Asc(e.KeyChar) <> 13 AndAlso Asc(e.KeyChar) <> 8 AndAlso Not
232         IsNumeric(e.KeyChar) Then
233         MessageBox.Show("Please use RFID Scanner")
234         e.Handled = True
235     End If
236 End Sub
237
238
239 Private Sub MetroTextBox9_TextChanged(sender As Object, e As EventArgs)
240     Handles MetroTextBox9.TextChanged
241     If MetroTextBox9.Text = "" Then
242         MetroTextBox10.Text = ""
243         MetroTextBox11.Text = ""
244     Else
245         Try
246             If objcon.State = ConnectionState.Closed Then objcon.Open()
247             com = New OleDb.OleDbCommand("Select B_ID,B_Name,B_Type from
248                 Borrowers", objcon)

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239         dr = com.ExecuteReader
240         While dr.Read
241             If dr.Item(0) = MetroTextBox9.Text Then
242                 MetroTextBox10.Text = dr.Item(1)
243                 MetroTextBox11.Text = dr.Item(2)
244             End If
245
246         End While
247         dr.Close()
248         objcon.Close()
249     Catch
250
251     End Try
252
253 End If
254 End Sub
255
256
257
258 Private Sub metrobutton4_Click_1(sender As Object, e As EventArgs) Handles ↗
259     MetroButton4.Click
260     LibraryMainForm.Show()
261     Me.Close()
262     SerialPort2.Close()
263 End Sub
264
265 Private Sub metrobutton2_Click_1(sender As Object, e As EventArgs) Handles ↗
266     MetroButton2.Click
267
268     If objcon.State = ConnectionState.Closed Then objcon.Open()
269
270     com = New OleDb.OleDbCommand("SELECT * FROM Issue", objcon)
271     com.Connection = objcon
272     Dim maxid As Object
273     Dim Strid As String
274     Dim intID As Integer
275
276     com.CommandText = "Select Max(Trans) as maxid from Issue"
277     maxid = com.ExecuteScalar
278
279     If maxid Is DBNull.Value Then
280         intID = 1
281     Else
282         Strid = CType(maxid, String)
283         intID = CType(Strid, Integer)
284         intID = intID + 1
285
286     End If
287
288     MetroTextBox1.Text = intID
289     objcon.Close()
290
291
292
```

```

293
294     If MetroTextBox9.Text = "" Then
295         MsgBox("Please enter the RFID Scanner!", 0, "")
296     ElseIf MetroTextBox8.Text = "" Then
297         MsgBox("Please enter the Barcode Book!", 0, "")
298     ElseIf MetroTextBox12.Text = 1 Then
299         MsgBox("You cannot borrow the last book", 0, "")
300     ElseIf MetroTextBox9.Text <> "" And MetroTextBox10.Text = "" Then
301         MsgBox("Unidentified Borrower", 0, "")
302     Else
303
304         Try
305
306
307             '////////////////////////////////////
308
309             If MetroTextBox8.Text = 2 Then
310                 If objcon.State = ConnectionState.Closed Then objcon.Open ➤
311                     ()
312                     com = New OleDb.OleDbCommand("UPDATE Books SET ➤
313                         status='Borrowed' WHERE BookID='" & MetroTextBox8.Text & ➤
314                         "'", objcon)
315                     com.ExecuteNonQuery()
316                     objcon.Close()
317                     Call readData()
318             Else
319                 If objcon.State = ConnectionState.Closed Then objcon.Open ➤
320                     ()
321                 com = New OleDb.OleDbCommand("UPDATE Books SET ➤
322                     status='Available' WHERE BookID='" & MetroTextBox8.Text & ➤
323                     "'", objcon)
324                 com.ExecuteNonQuery()
325                 objcon.Close()
326                 Call readData()
327             If objcon.State = ConnectionState.Closed Then objcon.Open()
328             com = New OleDb.OleDbCommand("UPDATE Books SET Quantity='" & ➤
329                 MetroTextBox12.Text - 1 & "' WHERE BookID='" & ➤
330                 MetroTextBox8.Text & "'", objcon)
331             com.ExecuteNonQuery()
332             objcon.Close()
333             Call readData()
334             If objcon.State = ConnectionState.Closed Then objcon.Open()
335             com = New OleDb.OleDbCommand("INSERT INTO Issue VALUES('" & ➤
336                 MetroTextBox8.Text & "','" & MetroTextBox6.Text & "','" & ➤
337                 MetroTextBox2.Text & "','" & MetroTextBox9.Text & "','" & ➤
338                 MetroTextBox10.Text & "','" & MetroDateTime1.Text & "','" & ➤
339                 MetroDateTime2.Text & "','" & MetroTextBox1.Text & "')", ➤
340                 objcon)
341             com.ExecuteNonQuery()
342             MsgBox("Book has been Issued!", 0, "")
343             Call readData()
344             objcon.Close()
345             '////////////////////////////////////

```

```
336         MetroTextBox9.Text = ""
337         MetroTextBox10.Text = ""
338         MetroTextBox11.Text = ""
339         MetroTextBox8.Text = ""
340         MetroTextBox6.Text = ""
341         MetroTextBox2.Text = ""
342         MetroTextBox3.Text = ""
343         MetroTextBox4.Text = ""
344         MetroTextBox7.Text = ""
345         MetroTextBox5.Text = ""
346         MetroTextBox12.Text = ""
347         MetroDateTime2.MinDate = DateTime.Today
348     Catch ex As Exception
349         MsgBox(ex.Message, 0, "")
350     End Try
351 End If
352
353
354 End Sub
355
356 Private Sub metrobutton1_Click(sender As Object, e As EventArgs) Handles metrobutton1.Click
357     MetroTextBox9.Text = ""
358     MetroTextBox10.Text = ""
359     MetroTextBox11.Text = ""
360
361
362     MetroTextBox8.Text = ""
363     MetroTextBox6.Text = ""
364     MetroTextBox2.Text = ""
365     MetroTextBox3.Text = ""
366     MetroTextBox4.Text = ""
367     MetroTextBox7.Text = ""
368     MetroTextBox5.Text = ""
369     MetroTextBox12.Text = ""
370     MetroDateTime2.MinDate = DateTime.Today
371 End Sub
372
373
374
375
376 End Class
```